Supplementary Information: Optimizing Complex Driver Behavior Models: A Case Study with a Continuous-State Active Inference Model

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APPENDIX

A. A differentiable active inference model of human driving

While the setup of the active inference model of human collision avoidance [1] and its theoretical underpinning is described somewhere else, this will be a more focused analysis the considers the best way to approach training such a model using gradient descent. In the following, we describe the model components, which are used in the implementation of the particle based model.

- The observation probability $p(\boldsymbol{o}|\boldsymbol{s}, \boldsymbol{\theta}_o)$ is assumed to be gaussian after a bijective transformation $\mathcal{L}$ on $\boldsymbol{s}$ and $\boldsymbol{o}$ (e.g., use looming), with

$$p(\boldsymbol{o}|\boldsymbol{s}, \boldsymbol{\theta}_o) = \mathcal{N}\left(\mathcal{L}(\boldsymbol{o})|\mathcal{L}(\boldsymbol{s}), \boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)\right) . \tag{A.1}$$

  It is used at two points in the active inference model:

  - Updating the belief using Bayesian inference. Here, given a prior believe $q_A(\boldsymbol{s})$ represented as the set $\boldsymbol{S}_A = \boldsymbol{s}_{A,1:N}$, we want to generate the posterior believe $\boldsymbol{S}$. In general, this is achieved by the following process:

    1) We generate a KDE approximation of q_A, with

$$q_A(\boldsymbol{s}) \approx \frac{1}{N}\sum_n \mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{s}_{A,n}), b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right) \tag{A.2}$$

    Here, $\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)$ is the sample based covariance over $\mathcal{L}(\boldsymbol{s})$, which is easily differentiable in regard to $\boldsymbol{S}_A$. Meanwhile, b is a scaling factor (commonly based on Silverman's rule of thumb).

    2) We then analytically generate the posterior, with

$$\begin{aligned} q(\boldsymbol{s}) &\propto\!\propto \sum_n \frac{1}{N}\mathcal{N}\left(\mathcal{L}(\boldsymbol{o})|\mathcal{L}(\boldsymbol{s}), \boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)\right)\mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{s}_{A,n}), b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right) \\ &\propto\!\propto \sum_n \frac{1}{N}\mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{o}), \boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o)\right)\mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{s}_{A,n}), b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right) \\ &\propto\!\propto \sum_n w_n\mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_{\text{post}}\right) \end{aligned} \tag{A.3}$$

    and

$$\begin{aligned} \boldsymbol{\Sigma}_{\text{post}} &= \left(\boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o)^{-1} + \left(b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right)^{-1}\right)^{-1} \\ \boldsymbol{\mu}_n &= \boldsymbol{\Sigma}_{\text{post}}\left(\boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o)^{-1}\mathcal{L}(\boldsymbol{o}) + \left(b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right)^{-1}\mathcal{L}(\boldsymbol{s}_{A,n})\right) \\ w_n &= \frac{1}{N}\mathcal{N}\left(\mathcal{L}(\boldsymbol{o})|\mathcal{L}(\boldsymbol{s}_{A,n}), \boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o) + b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A)\right) \end{aligned} \tag{A.4}$$

    Here, in the second line, using $\boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o)$ instead of $\boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)$ is a necessary approximation to make this analytically solvabale (i.e., $\mathcal{N}\left(\boldsymbol{s}|\boldsymbol{o}, \boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)\right)$ would not be a normal distribution over $\boldsymbol{s}$, and therefore, the rule for multiplying two Normals would be inapplicable). It is based on the assumption that $b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S}_A) \succ \succ \boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)$, which means that the posterior distribution will likely be close to the given observation $\boldsymbol{o}$[1]. It can be clearly seen that $\boldsymbol{\Sigma}_{\text{post}}$, $\boldsymbol{\mu}_n$, and w_n are differentiable in regard to the initial inputs $\boldsymbol{S}_A$, $\boldsymbol{o}$, and $\boldsymbol{\theta}_o$ (as long as $\boldsymbol{\Sigma}(\boldsymbol{o}, \boldsymbol{\theta}_o)$ and $\mathcal{L}$ are also differentiable).

    3) We sample our new sample from the resulting gaussian multi mixture model. In principle, this would be problematic in regard to differentiability, as sampling the mixture to use is non-differentiable. We can solve this in a two step process:

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[1]Alternatively, to preserve the locality, we could use $\boldsymbol{\Sigma}(\boldsymbol{s}_{A,n}, \boldsymbol{\theta}_o)$ instead, but this would be a more complex option

* We generate a new sample $\boldsymbol{s}_n$ from every kernel of the posterior GMM using the reparameterization trick:

$$\boldsymbol{s}_n = \mathcal{L}^{-1}(\boldsymbol{\mu}_n + \boldsymbol{L}_{\text{post}}\boldsymbol{\varepsilon}_n)\,,\ \boldsymbol{L}_{\text{post}}{\boldsymbol{L}_{\text{post}}}^T = \boldsymbol{\Sigma}_{\text{post}}\,,\ \boldsymbol{\varepsilon}_n \sim \mathcal{N}(\boldsymbol{\varepsilon}|\boldsymbol{0}, \boldsymbol{I}) \tag{A.5}$$

  Importantly, to make this more stable, we use the same $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_n$ for every mixture component.
* We use a straight-through Gumbel approach to select from $\boldsymbol{s}_n$, based on the fact that selecting based on the likelihoods w_n is the same as sampling $g_n \sim \text{Gumbel}(0, 1)$ from the standard Gumbel distribution and then choosing

$$\boldsymbol{s} = \boldsymbol{s}_{\arg\max_n g_n + \ln w_n}\,. \tag{A.6}$$

  During the backward pass, we then make this differentiable by using the softmax instead:

$$\boldsymbol{s} = \sum_n \text{softmax}(g_n + \ln w_n)\,\boldsymbol{s}_n\,. \tag{A.7}$$

Consequently, each new sample $\boldsymbol{s}$ would be differentiable towards all the different $\boldsymbol{\Sigma}_{\text{post}}$, $\boldsymbol{\mu}_n$, and w_n. Consequently, we can assume that this can be described as a stochastic function

$$\boldsymbol{S} = f_{\text{update}}(\boldsymbol{S}_A, \boldsymbol{o}, \boldsymbol{\theta}_o) \tag{A.8}$$

which is differentiable in regard to each of its inputs.

– Calculating the expected free energy G. Given a timestep where we have a belief $\widetilde{q}(\boldsymbol{s})$ reflected as a set of samples $\boldsymbol{S} = \boldsymbol{s}_{1:N}$, we define the EFE for such a timestep as g with

$$\begin{aligned} -g(\widetilde{q}(\boldsymbol{s})) &= \underbrace{\mathbb{E}_{\widetilde{q}(\boldsymbol{s})}\mathbb{E}_{p(\boldsymbol{o}|\boldsymbol{s},\boldsymbol{\theta}_o)}\ln(p(\boldsymbol{o}|\boldsymbol{\theta}_p))}_{\text{Pragmatic value } g_{\text{prag}}} + \underbrace{\mathcal{H}(\mathbb{E}_{\widetilde{q}(\boldsymbol{s})}p(\boldsymbol{o}|\boldsymbol{s},\boldsymbol{\theta}_o)) - \mathbb{E}_{\widetilde{q}(\boldsymbol{s})}\mathcal{H}(p(\boldsymbol{o}|\boldsymbol{s},\boldsymbol{\theta}_o))}_{\text{Epistemic value } g_{\text{epist}}} \\ &= \underbrace{\frac{1}{N}\sum_n \ln(p(\boldsymbol{o}_n|\boldsymbol{\theta}_p))}_{\text{Pragmatic value } g_{\text{prag}}} + \underbrace{\frac{1}{N}\sum_m -\ln\left(\frac{1}{N}\sum_n \int p(\boldsymbol{o}_m|\boldsymbol{s},\boldsymbol{\theta}_o)q_n(\boldsymbol{s})\mathrm{d}\boldsymbol{s}\right) - \frac{1}{N}\sum_n \mathcal{H}(p(\boldsymbol{o}|\boldsymbol{s}_n,\boldsymbol{\theta}_o))}_{\text{Epistemic value } g_{\text{epist}}}\,, \end{aligned} \tag{A.9}$$

where $p(\boldsymbol{o}|\boldsymbol{\theta}_p)$ is the preference prior. We will proceed under the assumption that this is differentiable in regard to both $\boldsymbol{o}$ and $\boldsymbol{\theta}_p$. Meanwhile, the samples $\boldsymbol{o}_n \sim p(|\boldsymbol{s}_n) = \mathcal{N}\left(\mathcal{L}(\boldsymbol{o})|\mathcal{L}(\boldsymbol{s}_n), \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)\right)$ can be generated using the standard reparameterization approach of

$$\boldsymbol{o}_n = \mathcal{L}^{-1}\left(\mathcal{L}(\boldsymbol{s}_n) + \boldsymbol{L}_o\boldsymbol{\varepsilon}_n\right)\,,\ \boldsymbol{L}_o{\boldsymbol{L}_o}^T = \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)\,,\ \boldsymbol{\varepsilon}_n \sim \mathcal{N}(\boldsymbol{\varepsilon}|\boldsymbol{0}, \boldsymbol{I}) \tag{A.10}$$

so that they are differentiable in regard to $\boldsymbol{s}_n$ and $\boldsymbol{\theta}_o$. Furthermore, we again use the KDE approximation

$$\widetilde{q}(\boldsymbol{s}) = \frac{1}{N}\sum q_n(\boldsymbol{s}) = \frac{1}{N}\sum \mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{s}_n), b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S})\right)\,. \tag{A.11}$$

In consequence, the pragmatic value g_{prag} is then definitely differentiable in regard to $\boldsymbol{S}$, $\boldsymbol{\theta}_p$, and $\boldsymbol{\theta}_o$. Meanwhile, this is a little bit more complex for the epistemic value g_{epist}, but it can be shown that we can simplify[2]

$$\begin{aligned} \int p(\boldsymbol{o}_m|\boldsymbol{s},\boldsymbol{\theta}_o)q_n(\boldsymbol{s})\mathrm{d}\boldsymbol{s} &= \int \mathcal{N}\left(\mathcal{L}(\boldsymbol{o}_m)|\mathcal{L}(\boldsymbol{s}), \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)\right)\mathcal{N}\left(\mathcal{L}(\boldsymbol{s})|\mathcal{L}(\boldsymbol{s}_n), b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S})\right)\mathrm{d}\boldsymbol{s} \\ &= \mathcal{N}\left(\mathcal{L}(\boldsymbol{o}_m)|\mathcal{L}(\boldsymbol{s}_n), \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o) + b(N)^2\widehat{\boldsymbol{\Sigma}}(\boldsymbol{S})\right)\,. \end{aligned} \tag{A.12}$$

and (assuming that $\boldsymbol{s}, \boldsymbol{o} \in \mathbb{R}^d$)

$$\begin{aligned} \mathcal{H}\left(p(\boldsymbol{o}|\boldsymbol{s}_n,\boldsymbol{\theta}_o)\right) &= \mathcal{H}\left(\mathcal{N}\left(\mathcal{L}(\boldsymbol{o})|\mathcal{L}(\boldsymbol{s}_n), \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)\right)\right) \\ &= \frac{1}{2}\ln\left(|\boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)|\right) + \frac{d}{2}\left(\ln(2\pi) + 1\right)\,. \end{aligned} \tag{A.13}$$

Importantly, it must be noted that in both the terms from Equations (A.12) and (A.13) we would have to consider the Jacobian $J_{\mathcal{L}}$. However, as these terms are subtracted, this cancels out (i.e., the epistemic value is invariant under the transformation $\mathcal{L}$ if $\mathcal{L}$ a diffeomorphism). Consequently, we can clearly see that the epistemic value is also a clearly differentiable function depending on $\boldsymbol{O} = \boldsymbol{o}_{1:N}$, $\boldsymbol{S}$ and $\boldsymbol{\theta}_o$.
Therefore, we can again describe this as an stochastic function

$$g = f_{\text{EFE}}\left(\boldsymbol{S}, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n\right) = -f_{\text{prag}}\left(\boldsymbol{S}, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n\right) - f_{\text{epis}}\left(\boldsymbol{S}, \boldsymbol{\theta}_o, H_n\right) \tag{A.14}$$

[2]Here, it should technically be $\mathcal{N}\left(\boldsymbol{o}_m|\boldsymbol{s}, \boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)\right)$ instead of $\mathcal{N}\left(\boldsymbol{o}_m|\boldsymbol{s}, \boldsymbol{\Sigma}(\boldsymbol{s}_n, \boldsymbol{\theta}_o)\right)$, but this is necessary for tractability. This assumption is again reasonable if we assume that $\boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)$ is relatively constant, which we should be able to assume.

that is again differentiable in regard to each of its inputs.

- The state transition function $p(\boldsymbol{s}'|\boldsymbol{s},\boldsymbol{a},\boldsymbol{\theta}_s)$ is assumed to have the form

$$p(\boldsymbol{s}'|\boldsymbol{s},\boldsymbol{a},\boldsymbol{\theta}_s) = p_0(\boldsymbol{s}'|\boldsymbol{s},\boldsymbol{a},\boldsymbol{\theta}_s)\breve{p}_{\text{norm}}\left(\boldsymbol{s}'|\boldsymbol{\theta}_s\right)$$
$$= \underbrace{\delta(\boldsymbol{s}'^c - f_\Phi(\boldsymbol{s},\boldsymbol{a},\Delta t))\mathcal{N}\left(\boldsymbol{s}'^a|\boldsymbol{s}^a,\boldsymbol{\Sigma}_a(\boldsymbol{\theta}_s)\right)}_{p_0} \underbrace{\min\left\{p_{\text{norm}}(\boldsymbol{s}'|\boldsymbol{\theta}_s), |T|\left(\sum_{\tau\in T} p_{\text{norm}}(\breve{\boldsymbol{s}}_\tau|\boldsymbol{\theta}_s)^{-1}\right)^{-1}\right\}}_{\breve{p}_{\text{norm}}} \tag{A.15}$$

with $T = \{1, H_n\}$. Here, $\boldsymbol{s}^a$ corresponds to the control states of the surrounding agents, while $\boldsymbol{s}^c$ refers to the remaining kinematic states in $\boldsymbol{s}$. Meanwhile, f_Φ refers to the underlying dynamic bicycle model with the give time step size Δt, while p_{norm} is a normative likelihood of a given state. Lastly, we have

$$\breve{\boldsymbol{s}}_\tau = \{f_\Phi(\boldsymbol{s}', \boldsymbol{a}, \tau\Delta t), \breve{\boldsymbol{s}}'^a\}\ . \tag{A.16}$$

We then use this to update the agents belief samples, where given a set $\boldsymbol{S} = \boldsymbol{s}_{1:N}$, we generate the updated belief $\boldsymbol{S}' = \boldsymbol{s}'_{1:N}$, with

$$\boldsymbol{s}'_n \sim p(\boldsymbol{s}'_n|\boldsymbol{s}_n,\boldsymbol{a},\boldsymbol{\theta}_s)\,. \tag{A.17}$$

Given that $\breve{p}_{\text{norm}}$ is likely not-normal, this is done in multiple steps[3]:

1) We sample M_n candidates from the normal distribution, with

$$\boldsymbol{s}'_m = \left\{\boldsymbol{s}'^c_m, \boldsymbol{s}'^a_m\right\}\ ,\ \boldsymbol{s}'^c_m = f_\Phi(\boldsymbol{s},\boldsymbol{a},\Delta t)\,,\ \boldsymbol{s}'^a_m \sim \mathcal{N}\left(\boldsymbol{s}'^a|\boldsymbol{s}^a,\boldsymbol{\Sigma}_a(\boldsymbol{\theta}_s)\right)\ . \tag{A.18}$$

Here, we can use the reparameterization approach from Equation (A.10) to ensure that $\boldsymbol{s}'^a_m$ is differentiable in regard to $\boldsymbol{s}^a$ and $\boldsymbol{\theta}_s$ – assuming that $\boldsymbol{\Sigma}_a(\boldsymbol{\theta}_s)$ is differentiable itself. For $\boldsymbol{s}'^c_m$, we need to look closer at f_Φ. This is based on the underlying differential equation

$$\dot{\boldsymbol{s}}_c = \Phi(\boldsymbol{s},\boldsymbol{a})\,, \tag{A.19}$$

where we assume that Φ (a bicycle model) is differentiable. As f_Φ is to be assumed to be the use of a Runge-Kutta method (in particular, we typically use Heun's method, a second order approach), it can be easily shown that f_Φ will be differentiable as well. Consequentially, $\boldsymbol{s}'_m$ will be differentiable in regard to both $\boldsymbol{s}$ and $\boldsymbol{\theta}_s$.

2) We calculate for each sample

$$p_m = \breve{p}_{\text{norm}}(\boldsymbol{s}'_m|\boldsymbol{\theta}_s) = \min\left\{p_{\text{norm}}(\boldsymbol{s}'_m|\boldsymbol{\theta}_s), |T|\left(\sum_{\tau\in T} p_{\text{norm}}(\breve{\boldsymbol{s}}_{m,\tau}|\boldsymbol{\theta}_s)^{-1}\right)^{-1}\right\}\ . \tag{A.20}$$

Here, in standard machine learning, we can generally assume that $\min$ will be differentiable. Furthermore, it can be assumed that given the discussion about the dynamic model Φ above and Equation (A.16) that $\breve{\boldsymbol{s}}_{m,\tau}$ will be differentiable with regard to $\boldsymbol{s}'_m$ and $\boldsymbol{a}$. Consequently, as long as $p_{\text{norm}}(\boldsymbol{s}|\boldsymbol{\theta}_s)$ is differentiable in regard to both its inputs, that p_m will be differentiable in regard to both $\boldsymbol{s}'_m$ and $\boldsymbol{\theta}_s$.

3) We now sample $\boldsymbol{s}'$ from the candidates $\boldsymbol{s}'_m$ based on the likelihoods p_m. For this, we use the straight-through Gumbel approach (see Equations (A.6) and (A.7)) to preserve differentiability.

In principle, we can therefore again summarize this as a stochastic function

$$\boldsymbol{S}' = f_{\text{prop}}(\boldsymbol{S},\boldsymbol{a},\boldsymbol{\theta}_s,H_n) \tag{A.21}$$

that is differentiable in regard to all of its inputs (assuming that f_Φ is differentiable in regard to the rollout time.

- The last part of the model to consider is the policy selection, which is using the Cross Entropy Method [2]. Here, a policy $\boldsymbol{\pi} = \boldsymbol{a}_{0:H-1}$ is a sequence of actions. In principle, this is again a iterative procedure:

1) we generate M initial policies $\boldsymbol{\pi}^{(0)}_m$ from a normal distribution.

$$\hat{\boldsymbol{\pi}}^{(0)}_m \sim \mathcal{N}\left(\hat{\boldsymbol{\pi}}|\boldsymbol{\mu}_{\pi,0}(\boldsymbol{\theta}_\pi),\boldsymbol{\Sigma}_{\pi,0}(\boldsymbol{\theta}_\pi)\right) \tag{A.22}$$

Here, we assume that $\boldsymbol{\mu}_{\pi,0}$ and $\boldsymbol{\Sigma}_{\pi,0}$ are themselves differentiable in regard to $\boldsymbol{\theta}_\pi$, while we ensure that $\hat{\boldsymbol{\pi}}^{(0)}_m$ is differentiable in regard to them by using the reparameterization trick in Equation (A.10).

2) We apply the pedal constraints f_{pedal} to the sampled policy to get the actually applied one $\boldsymbol{\pi}$, with

$$\boldsymbol{a}_\tau = f_{\text{pedal}}(\hat{\boldsymbol{a}}_\tau, \boldsymbol{a}_{\tau-1})\,. \tag{A.23}$$

[3]For readability, we will discard the n subscript in the following terms.

We consequently can define

$$\boldsymbol{\pi}_m^{(0)} = F_{\text{pedal}}(\hat{\boldsymbol{\pi}}_m^{(0)}, \boldsymbol{a}_{-1}) \tag{A.24}$$

3) Starting from the current state $\boldsymbol{S}_0$, we can use the function f_{prop} to generate the predicted states $\widetilde{\boldsymbol{S}}_{m,1:H}^{(0)}$. This was shown to be differentiable in regard to both $\boldsymbol{S}_0$ and $\boldsymbol{\pi}_m^{(0)}$.
4) We then calculate the epistemic value

$$G_m^{(0)} = \sum_{\tau=1}^{H_{\max}} w_\tau f_{\text{EFE}}(\widetilde{\boldsymbol{S}}_{m,\tau}^{(0)}, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n)\,. \tag{A.25}$$

During the forward pass, we use

$$w_\tau = \begin{cases} 1 & \tau \leq H \\ 0 & \tau > H \end{cases}, \tag{A.26}$$

while during backwards pass, we ensure differentiability in regard to H by using

$$w_\tau = \sigma(1.5\,(\tau - H)) \tag{A.27}$$

5) We run this procedure for K iterations, where $k \in \{0, \ldots, K-1\}$, for which we need to define the initial value $\boldsymbol{\mu}_{\pi,k+1}$ and $\boldsymbol{\Sigma}_{\pi,k+1}$ based on the previous results $\hat{\boldsymbol{\pi}}_{1:M}^{(k)}$ and $G_{1:M}^{(k)}$. Originally, this was done by taking the weighted mean and variance from $\hat{\boldsymbol{\pi}}_{1:M}^{(k)}$, where we had the weight

$$w_m^{(k)} = \begin{cases} \frac{1}{\beta M} & G_m^{(k)} > Q_{G_{1:M}^{(k)}}(1-\beta) \\ 0 & \text{otherwise} \end{cases}, \tag{A.28}$$

with $\beta \in (0,1)$ being the top fraction chosen, and Q is the quantile function (which is differentiable). In principle, this would be discontinuous and non differentiable function. If we want to avoid we can decide to instead calculate $w_m^{(k)}$ by using the sigmoid function with temperature T:

$$w_m^{(k)} = \sigma\left(\frac{G_m^{(k)} - Q_{G_{1:M}^{(k)}}(1-\beta)}{T}\right) \tag{A.29}$$

If so chosen, the selection of $\boldsymbol{\mu}_{\pi,k+1}$ and $\boldsymbol{\Sigma}_{\pi,k+1}$ becomes differentiable, as weighted mean and variance are always differentiable.
6) In the end, we then select our final policy

$$\boldsymbol{\pi}_{\text{opt}} = \boldsymbol{\pi}_{\arg\max_m G_m^{(K)}}\,. \tag{A.30}$$

In principle, we can then therefore again combine this into one stochastic function

$$\boldsymbol{\pi}_{\text{opt}} = f_{\text{plan},H}(\boldsymbol{S}_0, \boldsymbol{a}_{-1}, \boldsymbol{\theta}_\pi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n, T) \tag{A.31}$$

which is differentiable in regard to all inputs.

- In our model, to reproduce realistic response times, the full policy is not generated at each timestep. Instead, we first calculate a reference policy to estimate accumulated surprise, given a previous policy $\boldsymbol{\pi}_{-1}^*$:
  1) We take all but the first timestep of $\boldsymbol{\pi}_{-1}^*$ as inputs in iterative use of f_{prop} to advance $\boldsymbol{S}_0$ to $\widetilde{\boldsymbol{S}}_{H-1}$. We then generate the optimal final action as

$$\boldsymbol{a}_{H-1,\text{opt}} = f_{\text{plan},1}(\widetilde{\boldsymbol{S}}_{H-1}, \boldsymbol{a}_{H-2}^*, \boldsymbol{\theta}_\pi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n, T)\,. \tag{A.32}$$

  This is, as discussed above, differentiable.
  2) We construct the reference policy

$$\boldsymbol{\pi}_{\text{ref}} = \left(\boldsymbol{\pi}_{-1}^* \setminus \{\boldsymbol{a}_{-1}^*\}\right) \cup \{\boldsymbol{a}_{H-1,\text{opt}}\}\,. \tag{A.33}$$

  3) We calculate the surprise value. Under the assumption that $\max_{\boldsymbol{S}} f_{\text{prag}}(\boldsymbol{S}, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p) = 0$[4], this is calculated as

$$\varepsilon = -\sum_{\tau=1}^{H_{\max}} w_\tau f_{\text{prag}}(\widetilde{\boldsymbol{S}}_{\text{ref},\tau}, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n) > 0\,, \tag{A.34}$$

[4]We can always rescale $p(\boldsymbol{o}|\boldsymbol{\theta}_p)$ to ensure this upper limit.

where $\widetilde{\boldsymbol{S}}_{\text{ref}}$ refers to the predicted states resulting from applying the reference policy $\boldsymbol{\pi}_{\text{ref}}$, and w_τ is the same as in Equation (A.25).

4) Based on the previous surprise E^*_{-1}, we calculate the current surprise E_0 as

$$E_0 = E^*_{-1} + \lambda\varepsilon \tag{A.35}$$

Consequently, we can calculate this surprise using the function

$$\boldsymbol{\pi}_{\text{ref}}, E_0 = f_{\text{ref}}(\boldsymbol{\pi}^*_{-1}, E^*_{-1}, \boldsymbol{S}_0, \boldsymbol{\theta}_\pi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_o, \boldsymbol{\theta}_p, H_n, T, \lambda) \tag{A.36}$$

that is differentiable in regard to all of its inputs.

- In practice, the final policy selection would normally happen in the following way:

$$\begin{aligned} \boldsymbol{\pi}^* &= \mathbf{1}_{\text{replan}}\boldsymbol{\pi}_{\text{opt}} + (1 - \mathbf{1}_{\text{replan}})\boldsymbol{\pi}_{\text{ref}} \\ E^*_0 &= (1 - \mathbf{1}_{\text{replan}})E_0 \end{aligned} \tag{A.37}$$

with

$$\mathbf{1}_{\text{replan}} = E_0 \geq 1\,. \tag{A.38}$$

This is not differentiable. Therefore, to preserve differentiable, we use the following continuous replanning measure in the backwards pass instead:

$$\mathbf{1}_{\text{replan}} = \frac{1}{1 + \exp(-c(E_0 - 1))}\,. \tag{A.39}$$

Here, a higher value of c would approach the binary variant used originally. While this would require use to calculate $\boldsymbol{\pi}_{\text{opt}}$ at all times, which is not necessary during inference itself, our implementation of this model in Jax means that this is necessary anyways to preserve static shapes.

Consequentiality, the whole model simulation step can now be estimated as the function

$$\boldsymbol{\pi}^*, E^*_0, \boldsymbol{S}_0 = f_{\text{step}}(\boldsymbol{\pi}^*_{-1}, E^*_{-1}, \boldsymbol{S}_{-1}, \boldsymbol{o}, \boldsymbol{\Theta}) \tag{A.40}$$

that is differentiable in regard to all its inputs (including the model parameters $\boldsymbol{\Theta} = \{\boldsymbol{\theta}_o, \boldsymbol{\theta}_p, \boldsymbol{\theta}_s, \boldsymbol{\theta}_\pi, H_n, H, T, \lambda\}$), as long as the following model components can be assumed to be differentiable themselves:

- $\boldsymbol{\Sigma}(\boldsymbol{s}, \boldsymbol{\theta}_o)$: The variance in the observation probability. This is made differentiable with regard to the inputs by replacing the hard cutoff used by the looming threshold [1], [3] with a sigmoid function to interpolate between the pre and post threshold variances.
- $\mathcal{L}(\boldsymbol{s})$: The potential transformation needed from state space into the space where the observation probability is normally distributed. The same would be true for $\mathcal{L}^{-1}(\boldsymbol{s}_\mathcal{L})$, which needs to exist. This transformation is smooth and differentiable in the original model [1].
- $p(\boldsymbol{o}|\boldsymbol{\theta}_p)$: The preference prior used in the pragmatic value calculation. This is made smooth by replacing the binary conditions (e.g., overlap between vehicles in the collision cost or the violation/compliance with lane keeping) with sigmoid functions.
- $\boldsymbol{\Sigma}_a(\boldsymbol{\theta}_a)$: The variance in the control states of the surrounding vehicles. This is already differentiable.
- $\Phi(\boldsymbol{s}, \boldsymbol{a})$: The internal dynamic model, which we derive from the bicycle model. It is already differentiable.
- $p_{\text{norm}}(\boldsymbol{s}|\boldsymbol{\theta}_s)$: The normative likelihood of a specific state. This was made differentiable by replacing the binary lane/road keeping violations with sigmoid functions centered around the decision threshold.
- $\boldsymbol{\mu}_{\pi,0}(\boldsymbol{\theta}_\pi)$: The initial expected value of the first sampled policies. This is already differentiable.
- $\boldsymbol{\mu}_{\Sigma,0}(\boldsymbol{\theta}_\pi)$: The initial variance of the first sampled policies. This is already differentiable.
- $f_{\text{pedal}}(\hat{\boldsymbol{a}}, \boldsymbol{a})$: The pedal constraints imposed on the policy candidates in the cross entropy method. This is already differentiable.

Using f_{step}, we can make sure that the trajectory produced by the model will be differentiable in regard to the initial belief, policy, and accumulated surprise of the model as well as the underlying parameters $\boldsymbol{\Theta}$. Importantly, discrete hyperparameters such as N or M are definitely not differentiable, and would always have to be fitted, if so desired, in a outer zeroth-order loop. However, these parameters are only describing the model accuracy and therefore likely do not need to be fitted, as long as they are set sufficiently high.

Given that the number of trained parameters is relatively low, while the number of steps in the function evaluation (for instance, multiple CEM iterations requiring repeated belief propagation), we retrieve the gradients using forward-mode automatic differentiation [4]. Compared to the backpropagation – commonly used in machine learning – this approach is much less memory intensive for this type of active inference models.

B. Experimental Setup

This appendix will provide a in depth description of the experiment underlying our presented results.

TABLE I: An overview of the parameters that make up our parameter space $\mathcal{C}$. A more detailed description of those parameters can be found in the base model [1]. The signal to noise ratio shown here is for the model with parameters fixed to the initial values evaluated on a single steep lateral incursion scenario [5], and is averaged over all positions in the predicted trajectory. It is based on 128 repetitions. For more detail on the exact use of those parameters in the model, we refer readers to the original work by Schumann et al. [1].

Parameter	Initial	Interval	SNR	Description
Preference function $p(\boldsymbol{o})$				
$\frac{\mu_v}{v_0}$	1	$[0.9, 1.1]$	0.088	The relative mean of preference distribution of agent's velocity v compared to the initial speed.
σ_v $[ms^{-1}]$	0.5	$[0.25, 2.5]$	0.079	Standard deviation of preference distribution of agent's velocity v
σ_a $[ms^{-2}]$	0.2	$[0.1, 0.5]$	0.104	Standard deviation of preference distribution of agent's acceleration a
σ_ω $[s^{-1}]$	0.02	$[0.005, 0.1]$	0.092	Standard deviation of preference distribution of agent's steering rate ω
$-g_{LC}$	1000	$[250, 2000]$	0.111	Pragmatic cost for driving on a lane boundary or in opposing lanes
$-g_{LL}$	15000	$[10000, 45000]$	0.083	Pragmatic cost for leaving the road
$-g_C$	10000	$[8000, 15000]$	0.116	Pragmatic cost after having collided with a relative velocity of $10\,\mathrm{ms}^{-1}$
$-g_B$	5000	$[4000, 10000]$	0.097	Pragmatic cost for not keeping a sufficient distance to a lead vehicle
$-\sigma_\tau(\log_{10} g_C - 2)$ $[s^{-1}]$	0.05	$[0.02, 0.1]$	0.133	Relative standard deviation of preference distribution of looming coefficient $\tau = \max\{\frac{\dot{\varphi}}{\varphi} - 0.2\,\mathrm{s}^{-1}, 0\}$
State transition function $p(\boldsymbol{s}'\|\boldsymbol{s}, \boldsymbol{a})$				
$\sigma_{\omega,0}$ $[ms^{-2}]$	0.475	$[0.3, 1.0]$	0.072	Noise applied to other vehicle's steering rate during belief update
$\frac{\sigma_a}{\sigma_{a,0}} = \frac{\sigma_\omega}{\sigma_{\omega,0}}$	0.2	$[0.075, 0.25]$	0.092	The factor applied to the other's vehicle's action noise during behavior prediction
H_n	15	$[9, 15]$	0.049	Prediction horizon for the *projected normative probability* $\check{p}_{\text{norm}}$
$p_{n,\text{lane}}$	10^{-3}	$[10^{-4}, 2 \cdot 10^{-3}]$	0.077	The likelihood of lane violations in *normative probability* p_{norm}
$p_{n,\text{road}}$	10^{-5}	$[5 \cdot 10^{-6}, 10^{-4}]$	0.081	The likelihood of road violations in *normative probability* p_{norm}
Policy selection				
$\sigma_{a,CEM}$ $[ms^{-2}]$	5	$[2.5, 10]$	0.091	Noise for sampling accelerations a_τ in first CEM iteration
$\sigma_{\omega,CEM}$ $[s^{-1}]$	0.1	$[0.03, 0.1]$	0.110	Noise for sampling steering rate ω_τ in first CEM iteration
H	20	$[16, 24]$	0.089	The length of the considered policy (i.e., the agent's internal prediction horizon)

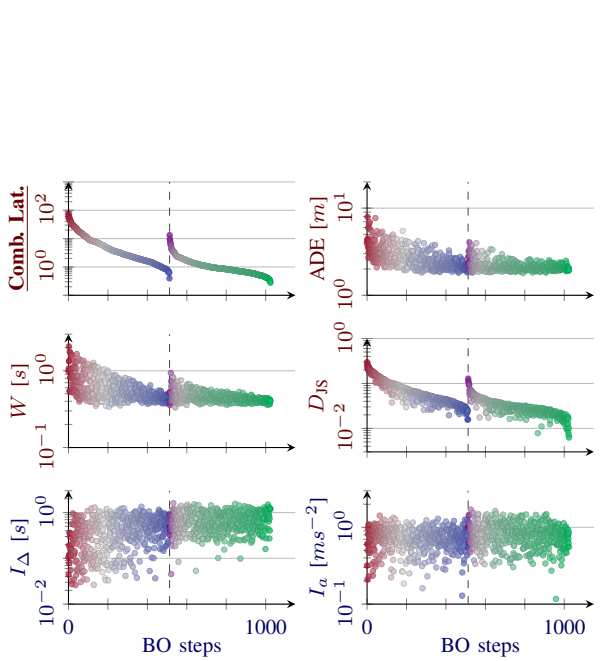
1) Optimizer hyperparameters: When running our experiments, we used the following hyperparameters for the optimizer setups.

a) Gradient Descent: We ran gradient descent with a batch size of $B = 8$ (i.e., how many samples where used for each parameter update). Given that the lateral incursion dataset [5] contains 64 samples, this results in 8 parameter updates per epoch, of which we run 150. For the gradient projection, we define the allowable projection space $\mathcal{C}$ as the same hyper-rectangle also used for the Bayesian optimization. For the consecutive use of the ADAM optimizer [6] (to which we pass those projected gradients), we used an initial learning rate of $\alpha = 0.01$, a decay rate of $\gamma = 0.975$ (applied once per epoch), and the exponential moment decay rates of $\beta_1 = 0.9$ and $\beta_2 = 0.995$ (applied per batch).

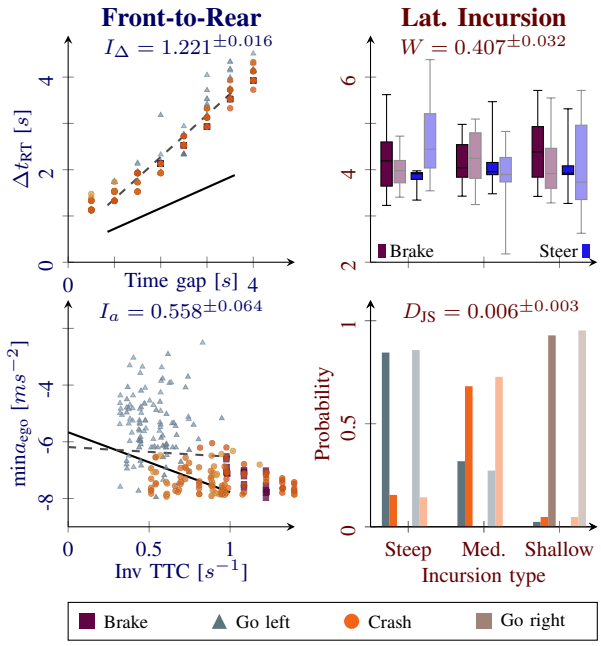
b) Bayesian Optimization: For the Bayesian optimization, we evaluated 512 initial samples from a scrambled Sobol sequence [7] from the valid parameter space $\mathcal{C}$, which we augmented by the hand-tuned parameter set from the original model [1]. These initial samples are needed to give a rough understanding of the objective function's behavior, and allow it to identify promising or underexplored regions of the search space. During the evaluation of these samples, we used bootstrapping with 512 repetitions to evaluate the noise inherent in the optimization objective by each time randomly selecting from the samples in the enlarged dataset. We then iteratively sampled new parameter candidates:

- We map the parameters, so that $\mathcal{C}$ would be mapped onto a hyper-cube.
- We scale all the previous objective values (and the corresponding noise), so that the variance of their expected values would be equal to 1. This should improve the conditioning of the surrogate model.
- We fit a Gaussian Process surrogate with a radial basis function kernel with automatic relevance determination to these data points, maximizing the marginal log likelihood using a L-BFGS-B optimizer [8].
- We use the *qLogNoisyExpectedImprovement* (qLogNEI) acquisition function to generate a single new parameters candidate. We speed up optimization by pruning the known points beforehand.
- We optimize qLogNEI by drawing the 256 initial samples, of which the 16 best are used as the starting point for optimization, again using a L-BFGS-B optimizer [8]. The point with the highest acquisition function value after this optimization is used as the next parameter candidate.
- We pass the candidate sample through the active inference model and the chosen objective, and add the result to the set of explored points.

2) Optimized parameters: The parameters we optimized in the work can be found in Table I. Specifically, with the exception of H and H_n, we optimize those parameters in logarithmic space, to better represent the influence of differences



(a) The objective values observed during Bayesian optimization. The different color schemes separate the initial sampling and the subsequent optimization process. The results during those two phases are sorted by the combined lateral incursion objective.



(b) The final results for the parameters minimizing the combined lateral incursion objective. The human data is represented by the solid black lines in the F2R scenario [9], [10] and the transparent elements in the lateral incursion scenario [5].

Fig. 1: The results of the Bayesian optimization using the combined lateral incursion objective (combining The Jensen Shannon divergence D_{JS} over the outcome distribution and the Wasserstein distance W over the response times), which is only taking into consideration the lateral incursion scenario. The color of the axis labels indicate the scenario underlying the corresponding objective.

across multiple orders of magnitude, and to avoid potential issues with negative values.

3) Combined objectives: We use the following combined objectives, where the factors were chosen so that the optimizer would try to improve upon all the underlying behavioral aspects in the original model [1] simultaneously:

- **Combined - Lateral Incursion**: This objective is defined as

$$\max\left\{\frac{W - 0.2\,\text{s}}{0.4\,\text{s}}, 0\right\}^2 + \max\left\{\frac{D_{\text{JS}} - 0.005}{0.035}, 0\right\}^2, \tag{A.41}$$

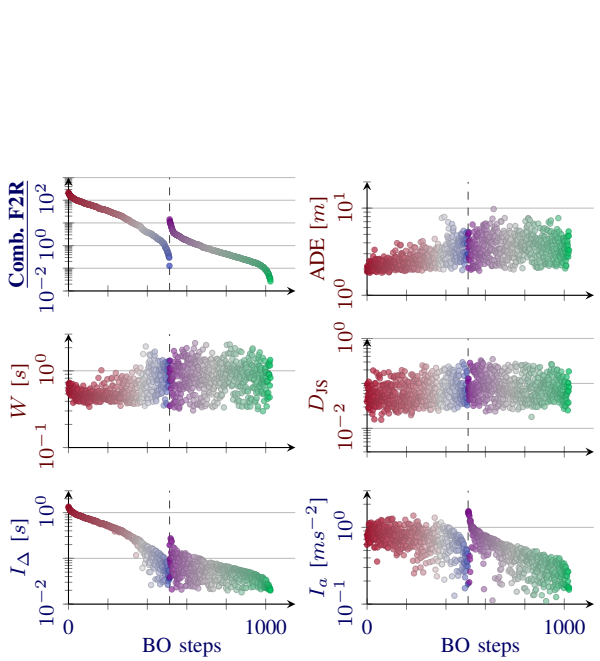
  where W is the Wasserstein distance between recorded and predicted response times and D_{JS} is the Jensen-Shannon divergence between the recorded and predicted outcomes, with both being averaged over the three lateral incursion severity settings in the dataset [5].
- **Combined - Front-to-Rear**: This objective is defined as

$$\max\left\{\frac{I_\Delta - 0.01\,\text{s}}{0.09\,\text{s}}, 0\right\}^2 + \max\left\{\frac{I_a - 0.3\,\text{m}\,\text{s}^{-2}}{0.4\,\text{m}\,\text{s}^{-2}}, 0\right\}^2, \tag{A.42}$$

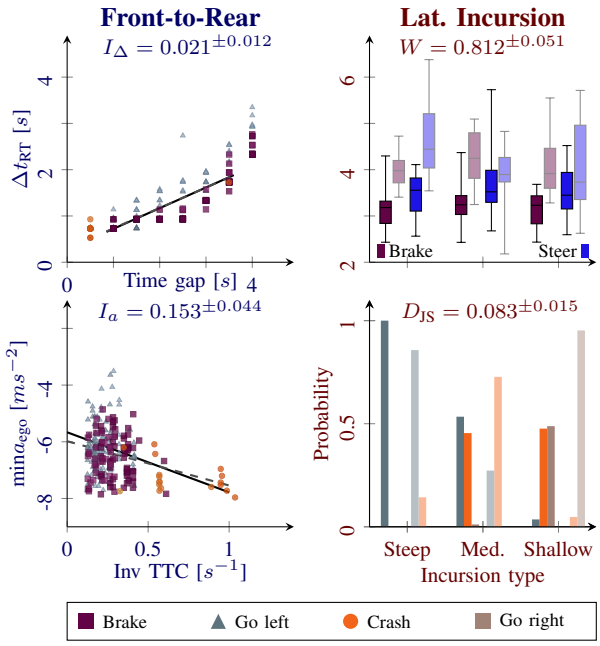
  where I_Δ and I_a are the average absolute expected errors [1] over the brake response time (conditioned on the initial time gap) and the minimum deceleration (conditioned on the inverse time-to-collision at the brake response), respectively.
- **Combined**: This is the sum of the other two combined objectives.

C. The problem of weakly constrained parameters for generalization

In general, generalizability is a desired aspect of any model of human driving. However, distribution shift is always a problem. This is especially crucial for mechanistic models, which might have parameters that can be weakly constrained by the training data. To investigate this issue, we conducted two additional experiments, where we evaluate performance on two distinct driving scenarios, woith only one being considered in the combined objective. First, we optimize the model's performance only on the lateral incursion scenario described by Johnson et al. [5] by matching both response times, quantified using the Wasserstein distance, and the collision avoidance outcome, quantified using the Jensen-Shannon divergence (JSD).



(a) The objective values observed during Bayesian optimization. The different color schemes separate the initial sampling and the subsequent optimization process. The results during those two phases are sorted by the combined front-to-rear objective.



(b) The final results for the parameters minimizing the combined front-to-rear objective. The human data is represented by the solid black lines in the F2R scenario [9], [10] and the transparent elements in the lateral incursion scenario [5].

Fig. 2: The results of the Bayesian optimization using the combined front-to-rear objective (combining the average absolute expected error over brake response time I_Δ and minium acceleration I_a), which is only taking into consideration the front-to-rear scenario. The color of the axis labels indicate the scenario underlying the corresponding objective.

Second, we optimized the model to simultaneously reproduce the brake response times reported for in a car-following (Front-to-Rear; F2R) scenario [9], as well as the magnitude of the resulting deceleration [10].

The results indicate a substantial sensitivity of model calibration to the scenarios included in the optimization objective. Specifically, optimization targeting only a single scenario tends to favor performance on that scenario at the expense of generalization to the other. For example, the model optimized exclusively for the lateral incursion scenarios performs substantially worse in the F2R scenario (Figure 1), while the converse is observed for the model optimized exclusively for the F2R scenario (Figure 2). This trade-off is particularly pronounced for response times. When the optimization is not constrained by both scenario types, the model tends to respond roughly half a second too quickly in the lateral incursion scenarios when calibrated to reproduce human-like response times in the F2R scenario. Conversely, optimization for the lateral incursion scenarios results in response times that are more than a second too slow in the F2R scenario. While optimizing each scenario individually can yield better performance on its respective objective compared to a joint optimization (Main paper Figure 4) – for instance, reducing D_{JS} from 0.17 to 0.006 – the magnitude of these improvements is often not statistically significant compared to the variability introduced by model noise; for example, the brake response-time fits in the F2R scenario differ by less than one standard deviation between the results of the two optimization runs where this scenario was considered. In other cases, the joint optimization results in little to no degradation. For instance, the Wasserstein distance of the response times in the lateral incursion scenarios becomes marginally worse under the combined objective, which may reflect a trade-off for the substantially improved performance in the F2R scenario.

These results are confirming potential fears about weakly constrained parameters. For instance, the parameter g_B (see Table I), which enforces an appropriate following distance, is expected to have little influence in a lateral-incursion scenario in which there is no lead vehicle that the ego vehicle needs to follow. When optimization is performed exclusively on the lateral-incursion data, this parameter is consequently only weakly constrained and may effectively be selected randomly. The resulting value is then likely poorly suited to the F2R scenario, where following behavior makes g_B substantially more important. The converse can occur for parameters that are primarily expressed in the lateral-incursion scenarios (such as the cost of leaving the road g_{LL}, which has only minor influence on policy selection in the F2R scenario but, by increasingly removing escape trajectories that come close to the road boundary, has a significant impact on collision outcomes in the lateral-incursion scenario). This scenario-specific conflict is particularly apparent for response times: the significant negative correlation between the lateral-incursion response-time MAE and the F2R scenario's I_Δ brake response-time error observed

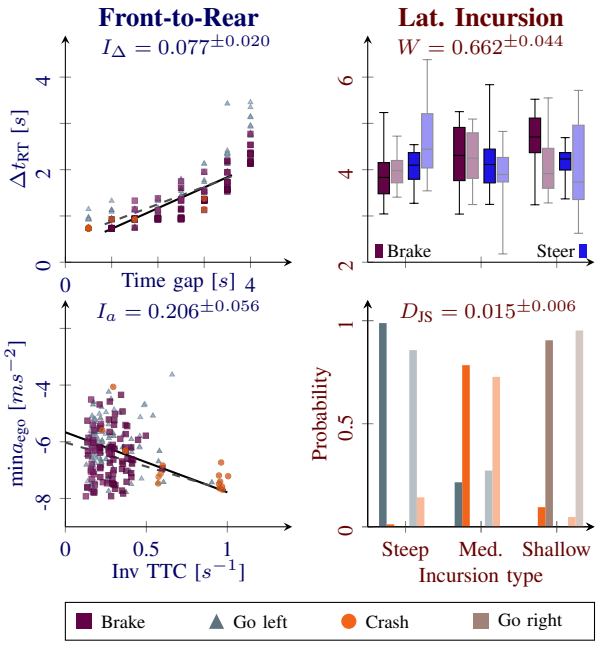


Fig. 3: The final results for the parameters minimizing the combined objective. The solid black lines in the F2R scenario represent the human data [9], [10], while the more transparent elements in the lateral incursion scenario represents the human data [5]. In this optimization run, the parameters of the state transition function in Table I remained fixed to the initial values.

in the main paper indicates a trade-off between reproducing response times in the two scenarios. However, these objectives become largely uncorrelated when considering only the best-performing subset of parameterizations, suggesting that this conflict is particularly pronounced for less well-calibrated models. The models optimized exclusively on a single dataset are likely representative of such parameterizations, as parameters that are weakly constrained by the optimization objective can result in poor performance on the omitted scenario. Thus, the apparent overfitting is not necessarily a failure of the model's general representational capability or the chosen optimization method, but rather a consequence of incomplete parameter identifiability when only a subset of the behavioral situations is used for calibration.

D. Implications of Hybrid Model Representations

An important problem in the field of mechanistic modeling is the use of hybrid model (using purely data-driven components such as neural networks). For such models, as question is whether all components of the generative model need to be calibrated jointly. One possible alternative would be a hybrid architecture in which for instance the state-transition function is replaced by a separately trained machine-learning model, while the active-inference parameters governing preferences and decision making remain subject to optimization. Such an approach could potentially reduce the number of parameters that need to be estimated and make use of the strong predictive capabilities of large deep-learning models.

However, this introduces an important risk of objective mismatch [11]. A model trained to reproduce observed physical trajectories is optimized to approximate the dynamics of the real environment, whereas the state-transition model in an active-inference agent also serves as part of the agent's internal generative model from which appropriate decisions must be derived. These objectives are related but not identical. In particular, a dynamics model that performs well on average may not provide an appropriate representation of uncertainty for the decision-making problem. If a learned model is trained predominantly on benign situations, for example, it may systematically underestimate uncertainty or assign excessive confidence to predictions in rare edge cases. Alternatively, a model learned on actual trajectories might misrepresent biases in a human's generative model that could be the cause of those behaviors. Such biases could subsequently affect planning and action selection even if the learned state-transition model has excellent conventional prediction capabilities.

Besides this objective mismatch issue, using fixed parameters for the state transition function can limit the expressivity of the overall model. As shown in Figure 3, fixing the state-transition function (even with hand-tuned parameters chosen with the role of being an agent's internal belief about world dynamics in mind), the optimization with a fixed state transition function results in a worse fit to human data, with the response-time objectives suffering a particularly sharp drop compared to the results from the optimization over all parameters (Main paper Figure 4). This is specifically noteworthy, as we did not change the number of evaluations between the optimization runs, which should have provided a much denser coverage

of the search space and therefore a higher likelihood to find the global optimum therein. Consequently, this indicates that tuning the generative model itself is crucial for giving the the model sufficient expressiveness to capture human behavior. Therefore, replacing the generative dynamics with a separately trained model (even ignoring potential objective mismatches in the training thereof) should not be regarded as a straightforward simplification: the learned representation could conflict with the needs of the downstream decision-making. Nevertheless, it is important to note that using say a deep learning model as an component of prediction model does not necessarily stand in contrast to those findings, as long as certain (hyper-) parameters of the model (e.g., the temperature in a softmax layer) are left tunable during the optimization. This would avoid the issue of a fixed component limiting expressiveness, as well as allowing for the correction of potential biases.

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